

Subject Name:- Mathematics For Computing.

MID-II Question Bank.

UNIT-III

Short Answer Questions.

1. Definition of Independent events.

Ans:- Independent events:- If E_1, E_2 are Independent events of sample space 'S' then $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

If E_1, E_2, E_3 are Independent events. then,

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3).$$

2. Definition of probability and write Axiom of probabilities?

Ans:- Definition of probability:- Let n be a mutually exclusive, equally likely and exhaustive events of random experiment. Let E be an event, m be number of favourable cases of an event E then probability of defined as,

$$P(E) = \frac{\text{number of favourable cases of } E}{\text{Total number of Events (or) Exhaustive Events.}}$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{m}{n}$$

now, E^c (or) $\bar{E}$ represents non-occurrence of E

number of favourable cases of $\bar{E} = n - m$.

$$\begin{aligned} \therefore P(\bar{E}) &= \frac{n(\bar{E})}{n(S)} = \frac{n-m}{n} = \frac{n}{n} - \frac{m}{n} \\ &= 1 - \frac{m}{n} \\ &= 1 - P(E) \end{aligned}$$

$$\therefore P(\bar{E}) = 1 - P(E)$$

$$\boxed{\therefore P(\bar{E}) + P(E) = 1}$$

Axioms of probability:- If all the Axioms are satisfied:

- (i) Axiom of positivity $P(E) \geq 0$
- (ii) Axiom of certainty $P(S) = 1$
- (iii) Axiom of union, If E_1, E_2 mutually exclusive events.

$$(i.e., E_1 \cap E_2 = \emptyset) \quad P(E_1 \cup E_2) = P(E_1) + P(E_2).$$

(3) write down multiplication theorem of probability.
Q82 If E_1, E_2 are events of sample space S , from Conditional probability,

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)}$$

$$i.e., P(E_1 \cap E_2) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right).$$

$$\text{and } P(E_1 \cap E_2) = P(E_2) \cdot P\left(\frac{E_1}{E_2}\right).$$

If E_1, E_2, E_3 are events of sample spaces, from Conditional probability,

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P\left(\frac{E_2}{E_1}\right) \cdot P\left(\frac{E_3}{E_1 \cap E_2}\right).$$

(4) Determine the probability for each of the following events. A non-defective bolt will be found it out of 600 bolts already examined 12 were defective.

Sol:-

$$\begin{aligned} \text{The probability of defective bolt, } P(D) &= \frac{12}{600} \\ &= \frac{1}{50}. \end{aligned}$$

$\therefore$ The probability of finding a non-defective bolt.

$$\therefore P(\bar{D}) = 1 - P(D) = 1 - \frac{1}{50} = \frac{50-1}{50} = \frac{49}{50}.$$

UNIT-III

Long Answer Questions

1. In a group there are 3 men and 2 women three persons are selected at random from this group. Find the probability that 1 man and 2 women (or) 2 men and 1 woman are selected.

Sol:-

Let S be the sample space associated with the selection of 3 persons out of 5.

$$\therefore n(S) = {}^5C_3 = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10.$$

Let E_1 be the event of selecting 1 man and 2 women.

$$n(E_1) = {}^3C_1 \times {}^2C_2 = 3 \times 1 = 3.$$

Let E_2 be the event of selecting 2 men and 1 woman.

$$n(E_2) = {}^3C_2 \times {}^2C_1 = \frac{3 \times 2}{2 \times 1} \times 2 = 6.$$

$$\text{But } E_1 \cap E_2 = \phi.$$

i.e., E_1, E_2 are mutually exclusive events.

Now, $E_1 \cup E_2$ is the event of selecting 1 man and 2 women or 2 men and 1 woman.

$$\begin{aligned}\therefore P(E_1 \cup E_2) &= P(E_1) + P(E_2) = \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} \\ &= \frac{3}{10} + \frac{6}{10} = \frac{9}{10}\end{aligned}$$

- (2) From a city, 3 newspapers A, B, C are published. A is read by 20%, B is read by 16%, C is read by 14%, Both A and B are read by 8%, Both A and C are read by 5%, Both B and C are read by 4%, and all A, B, C are read by 2%, what is the percentage of the population that need at least read one paper.

Sol:- Given that probability of newspaper read by A is 20%. i.e., $P(A) = \frac{20}{100}$.
and probability of newspaper read by B is 16%.
i.e., $P(B) = \frac{16}{100}$.

and probability of newspaper read by C is 14%

$$\text{i.e., } P(C) = \frac{14}{100}$$

probability of newspaper read by A and B is 8%

$$\text{i.e., } P(A \cap B) = \frac{8}{100}$$

probability of newspaper read by A and C is 5%

$$\text{i.e., } P(A \cap C) = \frac{5}{100}$$

probability of newspaper read by B and C is 4%

$$\text{i.e., } P(B \cap C) = \frac{4}{100}$$

probability of newspaper read by all A, B, C is 2%

$$\text{i.e., } P(A \cap B \cap C) = \frac{2}{100}$$

By addition theorem:-

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$= \frac{20}{100} + \frac{16}{100} + \frac{14}{100} - \frac{8}{100} - \frac{4}{100} - \frac{5}{100} + \frac{2}{100}$$

$$= \frac{20 + 16 + 14 - 8 - 4 - 5 + 2}{100}$$

$$= \frac{52-17}{100} = \frac{35}{100}$$

- ③ Find, (i) $P(A \cup B)$, (ii) $P(A^c \cap B)$, (iii) $P(A \cap B^c)$,
(iv) $P(A^c \cap B^c)$; If $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{5}$.

Sol:-

Given that, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{5}$.

(i) To find $P(A \cup B)$:-

$$\text{now, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{5}$$

$$= \frac{15+10-6}{30}$$

$$= \frac{19}{30}$$

$$\begin{array}{r|l} 2 & 2, 3, 5 \\ 3 & 1, 3, 5 \\ 5 & 1, 1, 5 \\ & 1, 1, 1 \end{array}$$

$$2 \times 3 \times 5 = 30$$

(ii) To find $P(A^c \cap B)$:-

$$\text{now, } P(A^c \cap B) = P(B) - P(A \cap B)$$

$$= \frac{1}{3} - \frac{1}{5}$$

$$= \frac{5-3}{15} = \frac{2}{15}$$

(iii) To find $P(A \cap B^c)$:-

$$\begin{aligned}\text{Now, } P(A \cap B^c) &= P(A) - P(A \cap B) \\ &= \frac{1}{2} - \frac{1}{5} \\ &= \frac{5-2}{10} \\ &= \frac{3}{10}\end{aligned}$$

(iv) To find $P(A^c \cap B^c)$:-

$$\begin{aligned}\text{Now, } P(A^c \cap B^c) &= P(A \cup B)^c \quad [\because P(A^c) = 1 - P(A)] \\ &= 1 - P(A \cup B) \\ &= 1 - \frac{19}{30} \\ &= \frac{30-19}{30} = \frac{11}{30}\end{aligned}$$

- (4) If 3 machines 1, 2, 3 produce 40%, 30%, 30% of total number of items of factory. The percentage of defective items of these machines are 4%, 2%, 3%. If an item is selected at random. Find the probability that the item is defective.

Sol:-

Let A, B, C be the events of item produced by machines 1, 2, 3.

Let D be defective items.

Probability that items produced by $P(A) = 40\% = \frac{40}{100}$.

$$\text{and } P(B) = \frac{30}{100}.$$

$$\text{and } P(C) = \frac{30}{100}.$$

and the percentage of defective items $P(D/A) = 4\%$,
 $= \frac{4}{100}$.

$$\text{and } P(D/B) = 2\%$$

$$= \frac{2}{100}.$$

$$\text{and } P(D/C) = 3\% = \frac{3}{100}.$$

The probability to chosen item is defective.

$$P(D) = P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)$$

$$= \frac{40}{100} \times \frac{4}{100} + \frac{30}{100} \times \frac{2}{100} + \frac{30}{100} \times \frac{3}{100}$$

$$= \frac{160}{10000} + \frac{60}{10000} + \frac{90}{10000}$$

$$= \frac{160+60+90}{10000} = \frac{310}{10000} = \frac{31}{1000}.$$

5

If A and B throw alternatively a pair of ordinary dice. A wins if he throws 6 before B throws 7 and if B wins if he throws 7 before A throws 6. If A begins show that his chance of winning is $\frac{30}{61}$.

Sol:-

Given that, $P(A) = \frac{5}{36}$. $\left[\because \text{Two dice} = 6 \times 6 = 36 \right]$

$$\text{Now, } P(\bar{A}) = 1 - P(A) = 1 - \frac{5}{36} = \frac{36-5}{36} = \frac{31}{36}$$

$$\text{and } P(B) = \frac{6}{36} = \frac{1}{6}$$

$$P(\bar{B}) = 1 - P(B) = 1 - \frac{1}{6} = \frac{6-1}{6} = \frac{5}{6}$$

$$\left[\because a + ar + ar^2 + \dots = \frac{a}{1-r} \right]$$

Chances of winning of A.

$$= P(A) + P(\bar{A}) \cdot P(\bar{B}) \cdot P(A) + P(\bar{A}) \cdot P(B) \cdot P(\bar{A}) \cdot P(\bar{B}) \cdot P(A) + \dots$$

$$= \frac{5}{36} + \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{5}{36} + \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{5}{36} + \dots$$

$$= \frac{5}{36} \left[1 + \frac{31}{36} \cdot \frac{5}{6} + \left(\frac{31}{36} \cdot \frac{5}{6} \right)^2 + \dots \right]$$

$$= \frac{5}{36} \left[\frac{1}{1 - \frac{31 \cdot 5}{36 \cdot 6}} \right]$$

$$= \frac{5}{36} \left[\frac{1}{1 - \frac{155}{216}} \right]$$

$$= \frac{5}{36} \left[\frac{1}{\frac{216 - 155}{216}} \right]$$

$$= \frac{5}{36} \left[\frac{1}{\frac{61}{216}} \right]$$

$$= \frac{5}{\cancel{36}_6} \left[\frac{\cancel{216}^{36}}{61} \right]$$

$$= \frac{5}{\cancel{6}} \left[\frac{\cancel{36}_6}{61} \right]$$

$$= \frac{30}{61}$$

unit-IV

Short Answer Questions.

- ① Define symmetric matrix and If $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ find symmetric matrix.

Sol:-

Definition of symmetric matrix:-

A square matrix $A = [a_{ij}]$ is said to be symmetric.
If $a_{ij} = a_{ji}$ for every i and j .

i.e., A is symmetric $\Leftrightarrow A = A^T$ or $A^T = A$.

Given that $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Find symmetric matrix:-

Now, $A^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

$\therefore \boxed{A = A^T}$

Given matrix is symmetric.

- ② Define Hermitian matrix and Find the Hermitian matrix, where $A = \begin{bmatrix} 4 & 1+3i \\ 1-3i & 7 \end{bmatrix}$

Sol:-

Definition of Hermitian matrix:-

A square matrix A such that $A^H = A$ is called Hermitian matrix.

$$\text{i.e., } A^H = (\bar{A})^T.$$

$$\therefore \boxed{A^H = (\bar{A})^T = A}$$

Given that, $A = \begin{bmatrix} 4 & 1+3i \\ 1-3i & 7 \end{bmatrix}$

To find Hermitian matrix:-

$$\text{Now, } \bar{A} = \begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$$

$$\text{Now, } (\bar{A})^T = \begin{bmatrix} 4 & 1+3i \\ 1-3i & 7 \end{bmatrix}$$

$$[\because A^H = (\bar{A})^T]$$

$$\therefore A^H = \begin{bmatrix} 4 & 1+3i \\ 1-3i & 7 \end{bmatrix} = A.$$

$$\therefore \boxed{A^H = A}$$

$\therefore A$ is Hermitian matrix.

(3)

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & -6 & x \end{bmatrix}$ and $|A| = 45$ find x ?

Sol:-

Given that, $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & -6 & x \end{bmatrix}$.

Given that $|A| = 45$.

Now, $\begin{vmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & -6 & x \end{vmatrix} = 45$

$1 \begin{vmatrix} 3 & 4 \\ -6 & x \end{vmatrix} - 0 \begin{vmatrix} 2 & 4 \\ 5 & x \end{vmatrix} + 0 \begin{vmatrix} 2 & 3 \\ 5 & -6 \end{vmatrix} = 45$

$1(3x + 24) - 0 + 0 = 45$ $\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right]$

$3x + 24 = 45$

$3x = 45 - 24$

$3x = 21$

$x = \frac{21}{3} = 7$

$\therefore \boxed{x = 7}$

④ Define join and meet. Find the join and meet of the zero-one matrices.

$$\text{If } A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Sol:-

Definition of join:-

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ matrices. Then the join of A and B is the zero-one matrix with $(i, j)^{\text{th}}$ entry $a_{ij} \vee b_{ij}$.

The join of A and B is denoted by $A \vee B$.

Definition of meet:-

The meet of A and B is the zero-one matrix with $(i, j)^{\text{th}}$ entry $a_{ij} \wedge b_{ij}$.

The meet of A and B is denoted by $A \wedge B$.

$$\text{Given that, } A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

To Find Join:-

Now, $A \vee B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

To find meet:-

Now, $A \wedge B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \wedge \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

⑤

Define trace of a matrix. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ find trace of A.

Sol:-

Definition of trace of a square matrix:-

Let $A = [a_{ij}]_{n \times n}$ then trace of the square matrix

A is defined as $\sum_{i=1}^n a_{ii}$ and is denoted by $\text{tr}(A)$.

$$\text{Thus } \text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}.$$

= sum of the principal diagonal elements of the square matrix A.

Given that, $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$.

Now, $\text{trace of } A = 1 + 3 + 3 = 7$.

(6)

Define singular and non-singular matrices.

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Singular matrix:- A square matrix A is said to be singular matrix if $|A| = 0$.

Non-singular matrix:- A square matrix A is said to be non-singular matrix if $|A| \neq 0$.

⑦ Define skew-symmetric matrix. show that the matrix (37)

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \text{ is skew-symmetric.}$$

Sol:-

Definition of skew-symmetric matrix:-

A square matrix $A = [a_{ij}]$ is said to be skew-symmetric

If $a_{ij} = -a_{ji}$ for every i and j .

i.e., A is skew symmetric $\Leftrightarrow A = -A^T$ (or) $A^T = -A$.

Given that, $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

$$\text{Now, } A^T = \begin{bmatrix} 0 & -1 \\ +1 & 0 \end{bmatrix}$$

$$= - \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = -A$$

$$\therefore \boxed{A^T = -A}$$

$\therefore A$ is skew-symmetric.

Hence proved.

⑧ Define skew-Hermitian matrix. To show that $A = \begin{bmatrix} 3i & 2+i \\ -2+i & -i \end{bmatrix}$

is skew-Hermitian.

Sol:-

Definition of skew-Hermitian Matrix:-

A square matrix 'A' such that $A^H = -A$ is called skew-Hermitian matrix.

$$\text{i.e., } A^H = (\bar{A})^T.$$

$$\therefore \boxed{A^H = -A}$$

$$\text{Given that, } A = \begin{bmatrix} -3i & 2+i \\ -2+i & -i \end{bmatrix}$$

$$\text{Now, } \bar{A} = \begin{bmatrix} 3i & 2-i \\ -2-i & i \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} 3i & -2-i \\ 2-i & i \end{bmatrix} \quad [\because A^H = (\bar{A})^T]$$

$$\therefore A^H = \begin{bmatrix} 3i & -2-i \\ 2-i & i \end{bmatrix}$$

$$= - \begin{bmatrix} -3i & 2+i \\ -2+i & -i \end{bmatrix} = -A.$$

$$\therefore A^H = -A.$$

$\therefore$ A is skew-Hermitian matrix.

Hence proved.

Unit-IV
Long Answer Questions.

(38)

①

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 6 & -4 & 2 \\ 3 & 4 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & -2 \\ 3 & 6 & -1 \end{bmatrix}$ then find $A+B$,
 $A-B$, $3B-2A$.

Sol:-

Given that, $A = \begin{bmatrix} 1 & 2 & 3 \\ 6 & -4 & 2 \\ 3 & 4 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & -2 \\ 3 & 6 & -1 \end{bmatrix}$

To find $A+B$:-

Consider, $A+B = \begin{bmatrix} 1 & 2 & 3 \\ 6 & -4 & 2 \\ 3 & 4 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & -2 \\ 3 & 6 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 1+1 & 2+6 & 3+(-2) \\ 6+3 & -4+4 & 2+(-2) \\ 3+3 & 4+6 & 6+(-1) \end{bmatrix} \quad \left[\begin{array}{l} \because +x-=- \\ -x+=- \\ -x-=- \\ +x+=+ \end{array} \right]$$

$$= \begin{bmatrix} 2 & 8 & 3-2 \\ 9 & 0 & 2-2 \\ 6 & 10 & 6-1 \end{bmatrix} = \begin{bmatrix} 2 & 8 & 1 \\ 9 & 0 & 0 \\ 6 & 10 & 5 \end{bmatrix}$$

$$\therefore A+B = \begin{bmatrix} 2 & 8 & 1 \\ 9 & 0 & 0 \\ 6 & 10 & 5 \end{bmatrix}$$

To find A-B:-

$$\text{Given, } A-B = \begin{bmatrix} 1 & 2 & 3 \\ 6 & -4 & 2 \\ 3 & 4 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & -2 \\ 3 & 6 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1-1 & 2-6 & 3-(-2) \\ 6-3 & -4-4 & 2-(-2) \\ 3-3 & 4-6 & 6-(-1) \end{bmatrix} \quad \left[\because -x = + \right]$$

$$= \begin{bmatrix} 0 & -4 & 3+2 \\ 3 & -8 & 2+2 \\ 0 & -2 & 6+1 \end{bmatrix}$$

$$\therefore A-B = \begin{bmatrix} 0 & -4 & 5 \\ 3 & -8 & 4 \\ 0 & -2 & 7 \end{bmatrix}$$

To find 3B-2A:-

$$\text{Now, } B = \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & -2 \\ 3 & 6 & -1 \end{bmatrix}$$

$$\text{Now, } 3B = 3 \begin{bmatrix} 1 & 6 & -2 \\ 3 & 4 & -2 \\ 3 & 6 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 18 & -6 \\ 9 & 12 & -6 \\ 9 & 18 & -3 \end{bmatrix}$$

$$\text{Now, } 2A = 2 \begin{bmatrix} 1 & 2 & 3 \\ 6 & -4 & 2 \\ 3 & 4 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & 6 \\ 12 & -8 & 4 \\ 6 & 8 & 12 \end{bmatrix}$$

$$\text{Consider, } 3B - 2A = \begin{bmatrix} 3 & 18 & -6 \\ 9 & 12 & -6 \\ 9 & 18 & -3 \end{bmatrix} - \begin{bmatrix} 2 & 4 & 6 \\ 12 & -8 & 4 \\ 6 & 8 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 3-2 & 18-4 & -6-6 \\ 9-12 & 12-(-8) & -6-4 \\ 9-6 & 18-8 & -3-12 \end{bmatrix}$$

$$[\because -x = +]$$

$$= \begin{bmatrix} 1 & 14 & -12 \\ -3 & 20 & -10 \\ 3 & 10 & -15 \end{bmatrix}$$

②

Solve the given equations by using cramer's Rule.

$$x+y+z=1, 2x+2y+3z=6; x+4y+9z=3.$$

Sol:-

$$\text{Given equations } x+y+z=1, 2x+2y+3z=6, x+4y+9z=3.$$

The matrix represent is $AX=D$.

$$\text{where, } A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \text{ and } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } D = \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix}.$$

Find Δ (or) $\det A$:-

$$\text{Now, } \Delta = \det A \text{ (or) } |A| = \begin{vmatrix} + & - & + \\ 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & 3 \\ 4 & 9 \end{vmatrix} - 1 \begin{vmatrix} 2 & 3 \\ 1 & 9 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix}.$$

$$[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc]$$

$$= 1(18-12) - 1(18-3) + 1(8-2)$$

$$= 6 - 15 + 6$$

$$= 12 - 15 = -3 \neq 0.$$

$$[\because \det A \neq 0].$$

To Find Δ_1 :-

$$\text{Now, } \Delta_1 = \begin{vmatrix} + & - & + \\ 1 & 1 & 1 \\ 6 & 2 & 3 \\ 3 & 4 & 9 \end{vmatrix}$$

$$\Delta_1 = 1 \begin{vmatrix} 2 & 3 \\ 4 & 9 \end{vmatrix} - 1 \begin{vmatrix} 6 & 3 \\ 3 & 9 \end{vmatrix} + 1 \begin{vmatrix} 6 & 2 \\ 3 & 4 \end{vmatrix}$$

$$\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right]$$

$$= 1(18-12) - 1(54-9) + 1(24-6)$$

$$= 6 - 45 + 18$$

$$= -21$$

To find Δ_2 :-

$$\text{Now, } \Delta_2 = \begin{vmatrix} + & - & + \\ 1 & 1 & 1 \\ 2 & 6 & 3 \\ 1 & 3 & 9 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 6 & 3 \\ 3 & 9 \end{vmatrix} - 1 \begin{vmatrix} 2 & 3 \\ 1 & 9 \end{vmatrix} + 1 \begin{vmatrix} 2 & 6 \\ 1 & 3 \end{vmatrix}$$

$$\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right]$$

$$= 1(54-9) - 1(18-3) + 1(6-6)$$

$$= 45 - 15 + 0$$

$$= 30$$

To find Δ_3 :-

$$\text{Now, } \Delta_3 = \begin{vmatrix} + & - & + \\ 1 & 1 & 1 \\ 2 & 2 & 6 \\ 1 & 4 & 3 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & 6 \\ 4 & 3 \end{vmatrix} - 1 \begin{vmatrix} 2 & 6 \\ 1 & 3 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix}.$$

$$\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right].$$

$$= 1(6 - 24) - 1(6 - 6) + 1(8 - 2)$$

$$= 1(-18) - 1(0) + 1(6).$$

$$= -18 + 6$$

$$= -12$$

Find x, y, z:-

$$\text{Now, } x = \frac{\Delta_1}{\Delta} = \frac{+21^7}{+3} = 7.$$

$$y = \frac{\Delta_2}{\Delta} = \frac{30^{10}}{-3} = -10.$$

$$z = \frac{\Delta_3}{\Delta} = \frac{+12^4}{+3} = 4.$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -10 \\ 4 \end{bmatrix}.$$

③

Solve the system of linear equations by Inverse Method, $x+y+z=1$, $2x+2y+3z=6$, $x+4y+9z=3$.

Sol:-

Given that, $x+y+z=1$, $2x+2y+3z=6$, $x+4y+9z=3$.

The matrix representation of $AX=B$.

$$\text{Now, } A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix}.$$

$$\text{Now, } AX=B.$$

$$X = A^{-1}B.$$

$$\text{where, } A^{-1} = \frac{\text{adj}A}{|A|}.$$

$$\text{where, } \text{adj}A = (\text{cofactors of } A)^T.$$

Find $|A|$ (or) $\det A$:-

$$\text{Now, } |A| = \det A = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & 3 \\ 4 & 9 \end{vmatrix} - 1 \begin{vmatrix} 2 & 3 \\ 1 & 9 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix}$$

$$\left[\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right]$$

$$|A| = 1(18-12) - 1(18-3) + 1(8-2)$$

$$= 6 - 15 + 6$$

$$= 12 - 15 = -3 \neq 0 \quad [\because \det A(x) |A| \neq 0]$$

Hence we can solve the given equations By using Matrix Inversion method.

$$\text{Now, } \text{adj} A = (\text{cofactors of } A)^T$$

Find cofactors of A:-

1st Row:-

$$\text{Given, } A = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$$

$$\text{cofactor of } 1 = (-1)^{i+j} = (-1)^{1+1} \begin{vmatrix} 2 & 3 \\ 4 & 9 \end{vmatrix} \quad \left[\begin{array}{l} \because i \text{ means row} \\ j \text{ means column} \end{array} \right]$$

$$= (-1)^2 [18-12] \quad \left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc \right]$$

$$= 6 \quad \left[\because (-1)^2 = -1 \times -1 = +1 \right]$$

$$\text{cofactor of } 1 = (-1)^{i+j} = (-1)^{1+2} \begin{vmatrix} 2 & 3 \\ 1 & 9 \end{vmatrix}$$

$$= (-1)^3 [18-3] \quad \left[\because (-1)^3 = -1 \times -1 \times -1 = -1 \right]$$

$$= -[15]$$

$$= -15$$

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$$\text{cofactor of } 1 = (-1)^{i+j} = (-1)^{1+3} \begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix}$$

$$= (-1)^4 [8-2] \quad [\because (-1)^4 = -1 \times -1 \times -1 \times -1 = +1]$$

$$= 6$$

2nd Row:-

$$\text{cofactor of } 2 = (-1)^{i+j} = (-1)^{2+1} \begin{vmatrix} 1 & 1 \\ 4 & 9 \end{vmatrix}$$

$$= (-1)^3 [9-4] \quad [\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc]$$

$$= -1 [5] \quad [\because (-1)^3 = -1 \times -1 \times -1 = -1]$$

$$= -5$$

$$\text{cofactor of } 2 = (-1)^{i+j} = (-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 4 & 9 \end{vmatrix}$$

$$= (-1)^4 [9-4] \quad [\because (-1)^4 = -1 \times -1 \times -1 \times -1 = +1]$$

$$= 1 [5]$$

$$= 5$$

$$\text{cofactor of } 3 = (-1)^{i+j} = (-1)^{2+3} \begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix}$$

$$= (-1)^5 [4-1] \quad [\because (-1)^5 = -1 \times -1 \times -1 \times -1 \times -1 = -1]$$

$$= -1 (3)$$

$$= -3$$

3rd Row:-

$$\begin{aligned}\text{cofactor of } 1 &= (-1)^{i+j} = (-1)^{3+1} \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} \\ &= (-1)^4 [3-2] \quad [\because (-1)^4 = -1 \times -1 \times -1 \times -1 = 1] \\ &= 1(1) \\ &= 1.\end{aligned}$$

$$\begin{aligned}\text{cofactor of } 4 &= (-1)^{i+j} = (-1)^{3+2} \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} \\ &= (-1)^5 [3-2] \quad [\because (-1)^5 = -1 \times -1 \times -1 \times -1 \times -1 = -1] \\ &= -1(1) \\ &= -1.\end{aligned}$$

$$\begin{aligned}\text{cofactor of } 9 &= (-1)^{i+j} = (-1)^{3+3} \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} \\ &= (-1)^6 [2-2] \\ &= 0.\end{aligned}$$

to find cofactor of A:-

$$\text{Now, cofactor of } A = \begin{bmatrix} 6 & -15 & 6 \\ -5 & 8 & -3 \\ 1 & -1 & 0 \end{bmatrix}$$

To find adjoint of A:-

$$\text{Now, adj } A = (\text{cofactor of } A)^T.$$

(13)

$$\text{Now, } \text{adj } A = \begin{bmatrix} 6 & -15 & 6 \\ -5 & 8 & -3 \\ 1 & -1 & 0 \end{bmatrix}^T$$

$$\therefore \text{adj } A = \begin{bmatrix} 6 & -5 & 1 \\ -15 & 8 & -1 \\ 6 & -3 & 0 \end{bmatrix}$$

To find A^{-1} :-

$$\text{Now, } A^{-1} = \frac{\text{adj } A}{|A|}$$

$$A^{-1} = \frac{\begin{bmatrix} 6 & -5 & 1 \\ -15 & 8 & -1 \\ 6 & -3 & 0 \end{bmatrix}}{-3}$$

$$\therefore A^{-1} = \frac{1}{-3} \begin{bmatrix} 6 & -5 & 1 \\ -15 & 8 & -1 \\ 6 & -3 & 0 \end{bmatrix}$$

To find x, y, z :-

$$\text{Now, } AX = B$$

$$X = A^{-1}B$$

$$X = \frac{1}{-3} \begin{bmatrix} 6 & -5 & 1 \\ -15 & 8 & -1 \\ 6 & -3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix}$$

$$X = \frac{1}{-3} \begin{bmatrix} 6-30+3 \\ -15+48-3 \\ 6-18+0 \end{bmatrix}$$

$$= \frac{1}{-3} \begin{bmatrix} 9-30 \\ -18+48 \\ 6-18 \end{bmatrix}$$

$$= \frac{1}{-3} \begin{bmatrix} -21 \\ 30 \\ -12 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{-21}{-3} \\ \frac{30}{-3} \\ \frac{-12}{-3} \end{bmatrix}$$

$$= \begin{bmatrix} 7 \\ -10 \\ 4 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -10 \\ 4 \end{bmatrix}$$

$$\therefore x=7, y=-10, z=4.$$

(4)

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then show that $A^2 - 4A - 5I = 0$. (44)

Sol:-

Given that $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$.

Show that $A^2 - 4A - 5I = 0$.

Now, $A^2 = A \times A$.

$$= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 1 + 2 \times 2 + 2 \times 2 & 1 \times 2 + 2 \times 1 + 2 \times 2 & 1 \times 2 + 2 \times 2 + 2 \times 1 \\ 2 \times 1 + 1 \times 2 + 2 \times 2 & 2 \times 2 + 1 \times 1 + 2 \times 2 & 2 \times 2 + 1 \times 2 + 2 \times 1 \\ 2 \times 1 + 2 \times 2 + 1 \times 2 & 2 \times 2 + 2 \times 1 + 1 \times 2 & 2 \times 2 + 2 \times 2 + 1 \times 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

Now, $4A = 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix}$

$$\text{and } 5I = 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Consider, $A^2 - 4A - 5I$.

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

$$= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore A^2 - 4A - 5I = 0$$

Hence proved.

⑤ Define orthogonal matrix. Show that $A = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$ is orthogonal.

Soln

orthogonal matrix

A square matrix A is said to be orthogonal matrix. If $AA^T = A^T A = I$.

$$\text{Given that, } A = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

$$\text{Now, } A^T = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

$$\text{Consider, } AA^T = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1+1+1+1 & -1-1+1+1 & -1+1-1+1 & -1+1+1-1 \\ -1+1+1+1 & 1+1+1+1 & 1-1+1+1 & 1-1+1-1 \\ -1+1+1+1 & 1-1+1+1 & 1+1+1+1 & 1+1-1-1 \\ -1+1+1+1 & 1-1+1-1 & 1+1-1-1 & 1+1+1+1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I.$$

$$\therefore AA^T = I.$$

$\therefore A$ is an orthogonal matrix.

(6)

Define matrix multiplication. If $A = \begin{bmatrix} -6 & 3 \\ 4 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 6 & -8 \end{bmatrix}$, find AB, BA, AA^T, BB^T .

Sol:-

Definition of Matrix Multiplication:-

Let $A = [a_{ik}]_{m \times n}$ and $B = [b_{kj}]_{n \times p}$ then the

matrix $C = [c_{ij}]_{m \times p}$, where $c_{ij} = \sum_{k=1}^n a_{ik} \cdot b_{kj}$

is called the product of the matrices A and B in that order. we can write $C = AB$.

Given that, $A = \begin{bmatrix} -6 & 3 \\ 4 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 6 & -8 \end{bmatrix}$.

To find AB :-

$$\text{Now, } AB = \begin{bmatrix} -6 & 3 \\ 4 & -7 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 6 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} (-6) \times 4 + 3 \times 6 & -6 \times 3 + 3 \times (-8) \\ 4 \times 4 + (-7) \times 6 & 4 \times 3 + (-7) \times (-8) \end{bmatrix}$$

$$AB = \begin{bmatrix} 24+18 & -18-24 \\ -16-42 & 12+56 \end{bmatrix}$$

$$= \begin{bmatrix} 42 & -42 \\ -58 & 68 \end{bmatrix}$$

To find BA:-

$$\text{Now, } BA = \begin{bmatrix} -4 & 3 \\ 6 & -8 \end{bmatrix} \begin{bmatrix} -6 & 3 \\ 4 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} (-4) \times (-6) + 3 \times 4 & -4 \times 3 + 3 \times (-7) \\ 6 \times (-6) + (-8) \times 4 & 6 \times 3 + (-8) \times (-7) \end{bmatrix}$$

$$BA = \begin{bmatrix} 24+12 & -12-21 \\ -36-32 & 18+56 \end{bmatrix}$$

$$BA = \begin{bmatrix} 36 & -33 \\ -68 & 74 \end{bmatrix}$$

To find AA^T :-

$$\text{Now, } A = \begin{bmatrix} -6 & 3 \\ 4 & -7 \end{bmatrix}$$

(47)

$$\text{and } A^T = \begin{bmatrix} -6 & 4 \\ 3 & -7 \end{bmatrix}.$$

$$\text{Consider, } AA^T = \begin{bmatrix} -6 & 3 \\ 4 & -7 \end{bmatrix} \begin{bmatrix} -6 & 4 \\ 3 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} -6 \times (-6) + 3 \times 3 & -6 \times 4 + 3 \times (-7) \\ 4 \times (-6) + (-7) \times 3 & 4 \times 4 + (-7) \times (-7) \end{bmatrix}$$

$$\text{Now, } AA^T = \begin{bmatrix} 36+9 & -24-21 \\ -24-21 & 16+49 \end{bmatrix}$$

$$\therefore AA^T = \begin{bmatrix} 45 & -45 \\ -45 & 65 \end{bmatrix}.$$

To find BB^T :-

$$\text{Now, } B = \begin{bmatrix} -4 & 3 \\ 6 & -8 \end{bmatrix}$$

$$\text{and } B^T = \begin{bmatrix} -4 & 6 \\ 3 & -8 \end{bmatrix}$$

Consider, $BB^T = \begin{bmatrix} -4 & 3 \\ 6 & -8 \end{bmatrix} \begin{bmatrix} -4 & 6 \\ 3 & -8 \end{bmatrix}$

$$= \begin{bmatrix} -4 \times (-4) + 3 \times 3 & -4 \times 6 + 3 \times (-8) \\ 6 \times (-4) + (-8) \times 3 & 6 \times 6 + (-8) \times (-8) \end{bmatrix}$$

now, $BB^T = \begin{bmatrix} 16+9 & -24-24 \\ -24-24 & 36+64 \end{bmatrix}$

$$\therefore BB^T = \begin{bmatrix} 25 & -48 \\ -48 & 100 \end{bmatrix}$$

(7)

If $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ then show that A is

Hermitian and iA is Skew-Hermitian.

Sol:-

Definition of Hermitian Matrix:-

A square matrix A such that $A^H = A$ is called Hermitian Matrix.

i.e., $A^H = (\bar{A})^T$

$\therefore A^H = A$

(48)

Given that, $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$

Show that A is Hermitian:-

Now, $A = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$

and, $\bar{A} = \begin{bmatrix} 3 & 7+4i & -2-5i \\ 7-4i & -2 & 3-i \\ -2+5i & 3+i & 4 \end{bmatrix}$

now, $(\bar{A})^T = \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$

$\therefore A^H = (\bar{A})^T = A \quad \left[\because A^H = (\bar{A})^T \right]$

$\therefore A^H = A$

$\therefore$ A is an Hermitian Matrix.
Hence proved.

Definition of skew-Hermitian Matrix:-

A square Matrix A such that $A^D = -A$ is called skew-Hermitian Matrix.

$$\text{i.e., } A^D = (\bar{A})^T = -A.$$

To show that iA is skew-Hermitian:-

$$\text{Let, } B = iA = i \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}.$$

$$= \begin{bmatrix} 3i & 7i-4i^2 & -2i+5i^2 \\ 7i+4i^2 & -2i & 3i+i^2 \\ -2i-5i^2 & 3i-i^2 & 4i \end{bmatrix} \quad [\because i^2 = -1].$$

$$= \begin{bmatrix} 3i & 7i-4(-1) & -2i+5(-1) \\ 7i+4(-1) & -2i & 3i+(-1) \\ -2i-5(-1) & 3i-(-1) & 4i \end{bmatrix}$$

$$= \begin{bmatrix} 3i & 7i+4 & -2i-5 \\ 7i-4 & -2i & 3i-1 \\ -2i+5 & 3i+1 & 4i \end{bmatrix}$$

(5)

$$\text{Now, } \overline{B} = \begin{bmatrix} -3i & -7i+4 & 2i-5 \\ -7i-4 & 2i & -3i-1 \\ 2i+5 & -3i+1 & -4i \end{bmatrix}$$

$$(\overline{B})^T = \begin{bmatrix} -3i & -7i-4 & 2i+5 \\ -7i+4 & 2i & -3i+1 \\ 2i-5 & -3i-1 & -4i \end{bmatrix}$$

$$B^{\theta} = \begin{bmatrix} -3i & -7i-4 & 2i+5 \\ -7i+4 & -2i & -3i+1 \\ 2i-5 & -3i-1 & -4i \end{bmatrix}$$

By the definition of skew-Hermitian matrix is,

$$B^{\theta} = -B.$$

$$\therefore B^{\theta} = (\overline{B})^T.$$

$$B^{\theta} = - \begin{bmatrix} 3i & 7i+4 & -2i-5 \\ 7i-4 & -2i & 3i-1 \\ -2i+5 & 3i+1 & 4i \end{bmatrix}$$

$$B^{\theta} = -B$$

$\therefore B = iA$ is skew-Hermitian matrix.

Hence proved.

(8)

Solve the given Linear equations by using Cramer's Rule,

$$x - y + 3z = 5, 4x + 2y - z = 0, -x + 3y + z = 5.$$

Sol:-

Given, equations are, $x - y + 3z = 5$, $4x + 2y - z = 0$,
 $-x + 3y + z = 5$.

The matrix representation is $AX = D$.

$$\text{where, } A = \begin{bmatrix} 1 & -1 & 3 \\ 4 & 2 & -1 \\ -1 & 3 & 1 \end{bmatrix} \text{ and } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } D = \begin{bmatrix} 5 \\ 0 \\ 5 \end{bmatrix}.$$

To find $\Delta = \det A$ (or) $|A|$:-

$$\text{Now, } \Delta = \det A = |A| = \begin{vmatrix} 1 & -1 & 3 \\ 4 & 2 & -1 \\ -1 & 3 & 1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} + 1 \begin{vmatrix} 4 & -1 \\ -1 & 1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 2 \\ -1 & 3 \end{vmatrix}$$

$$= 1(2 - (-3)) + 1(4 - 1) + 3(12 - (-2))$$

$$= 1(2+3) + 1(3) + 3(14) = 5 + 3 + 42 = 50.$$

$$\therefore \Delta = 50 \neq 0$$

$$[\because \det A \neq 0 \text{ (or) } |A| \neq 0]$$

To find Δ_1 :-

$$\text{now, } \Delta_1 = \begin{vmatrix} + & - & + \\ 5 & -1 & 3 \\ 0 & 2 & -1 \\ 5 & 3 & 1 \end{vmatrix}$$

$$[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc]$$

$$= 5 \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} + 1 \begin{vmatrix} 0 & -1 \\ 5 & 1 \end{vmatrix} + 3 \begin{vmatrix} 0 & 2 \\ 5 & 3 \end{vmatrix}$$

$$= 5(2 - (-3)) + 1(0 - (-5)) + 3(0 - 10)$$

$$= 5(2 + 3) + 1(5) + 3(-10)$$

$$= 5(5) + 5 - 30 = 25 + 5 - 30 = 30 - 30 = 0$$

$$\therefore \Delta_1 = 0$$

To find Δ_2 :-

$$\text{now, } \Delta_2 = \begin{vmatrix} + & - & + \\ 1 & 5 & 3 \\ 4 & 0 & -1 \\ -1 & 5 & 1 \end{vmatrix}$$

$$[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc]$$

$$= 1 \begin{vmatrix} 0 & -1 \\ 5 & 1 \end{vmatrix} - 5 \begin{vmatrix} 4 & -1 \\ -1 & 1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 0 \\ -1 & 5 \end{vmatrix}$$

$$= 1(0 - (-5)) - 5(4 - (-1)) + 3(20 - (0))$$

$$= 1(5) - 5(5) + 3(20) = 5 - 25 + 60 = 35 - 25 = 10$$

$$\therefore \Delta_2 = 10$$

To find Δ_3 :-

$$\text{Now, } \Delta_3 = \begin{vmatrix} + & - & + \\ 1 & -1 & 5 \\ 4 & 2 & 0 \\ -1 & 3 & 5 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & 0 \\ 3 & 5 \end{vmatrix} + 1 \begin{vmatrix} 4 & 0 \\ -1 & 5 \end{vmatrix} + 5 \begin{vmatrix} 4 & 2 \\ -1 & 3 \end{vmatrix} \quad \left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right]$$

$$= 1(10 - 0) + 1(20 + 0) + 5(12 - 2)$$

$$= 10 + 20 + 5(10)$$

$$= 10 + 20 + 50 = 100$$

$$\therefore \Delta_3 = 100$$

To find x, y, z :-

$$\text{Now, } x = \frac{\Delta_1}{\Delta} = \frac{0}{100} = 0$$

$$y = \frac{\Delta_2}{\Delta} = \frac{50}{50} = 1$$

$$z = \frac{\Delta_3}{\Delta} = \frac{100}{50} = 2$$

$$\therefore x=0, y=1, z=2$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$\underline{\underline{=}}$

(9)

Test whether the $T: V \rightarrow W$ is defined by

$$T(x, y, z) = (x-y, y-z).$$

Sol:-Definition Linear Transformation:-

Let V and W be vector spaces. A function $T: V \rightarrow W$ is called linear transformation. If and only if

$$(i) \quad T(\alpha + \beta) = T(\alpha) + T(\beta)$$

$$(ii) \quad T(c\alpha) = cT(\alpha).$$

If the given equation satisfies ~~to~~ these two conditions then given equation linear transformation.

$$\text{Given that, } T(x, y, z) = (x-y, y-z).$$

To prove T is a linear transformation:-

It is enough to prove (i) $T(\alpha + \beta) = T(\alpha) + T(\beta)$

$$(ii) \quad T(c\alpha) = cT(\alpha)$$

$$\text{Now, } \alpha = (x_1, y_1, z_1) = T(\alpha) = T(x_1, y_1, z_1) = (x_1 - y_1, y_1 - z_1).$$

$$\beta = (x_2, y_2, z_2) = T(\beta) = T(x_2, y_2, z_2) = (x_2 - y_2, y_2 - z_2).$$

TO PROVE
(i) $T(\alpha + \beta) = T(\alpha) + T(\beta)$:-

Consider, L.H.S :-

$$T(\alpha + \beta) = T\left(\left(x_1, y_1, z_1\right) + \left(x_2, y_2, z_2\right)\right)$$

$$= T\left(\underbrace{x_1 + x_2}_x, \underbrace{y_1 + y_2}_y, \underbrace{z_1 + z_2}_z\right)$$

$$= \left(\underbrace{x_1 + x_2}_{x} - \underbrace{y_1 + y_2}_{y}, \underbrace{y_1 + y_2}_{y} - \underbrace{z_1 + z_2}_{z}\right) \quad \left[\because T(x, y, z) = (x - y, y - z) \right]$$

$$= \left(x_1 + x_2 - y_1 - y_2, y_1 + y_2 - z_1 - z_2\right)$$

$$= \left(x_1 - y_1, y_1 - z_1\right) + \left(x_2 - y_2, y_2 - z_2\right)$$

$$\left[\because T(\alpha) = (x_1 - y_1, y_1 - z_1) \right]$$

$$\left[\because T(\beta) = (x_2 - y_2, y_2 - z_2) \right]$$

$$= T(\alpha) + T(\beta)$$

$$\therefore T(\alpha + \beta) = T(\alpha) + T(\beta)$$

(ii) To prove $T(ca) = cT(a)$:-

$$\text{Consider, } T(ca) = T(c(x, y, z))$$

$$= T(cx, cy, cz)$$

$$\left[\because T(\underbrace{cx}_x, \underbrace{cy}_y, \underbrace{cz}_z) \right]$$

$$\left[\because T(x, y, z) = (x-y, y-z) \right]$$

$$= (cx - cy, cy - cz)$$

$$= c(x-y, y-z) \quad \left[\because T(a) = (x-y, y-z) \right]$$

$$= cT(a)$$

$$\therefore \boxed{T(ca) = cT(a)}$$

Hence proved.

10

If A.P are 6, 11, 16 -----, Find T_{20} and S_{20} .

Sol:-

Given, A.P are 6, 11, 16 -----

Now Applying n^{th} term Formula:- $T_n = a + (n-1)d$

Given, $a=6$, $d=5$, $n=20$.

$$\left[\begin{array}{l} \because d=11-6=5 \\ d=16-11=5 \end{array} \right]$$

To Find T_{20} :-

$$T_n = a + (n-1)d$$

$$T_{20} = 6 + (20-1)5$$

$$= 6 + (19)5$$

$$= 6 + 95$$

$$= 101$$

$$\therefore \boxed{T_{20} = 101}$$

Sum of n terms formula:-

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

To Find S_{20} :-

Given, $a=6$, $d=5$, $n=20$.

$$\text{Now, } S_{20} = \frac{20}{2} [2(6) + (20-1)5]$$

$$= 10 [12 + 19 \times 5]$$

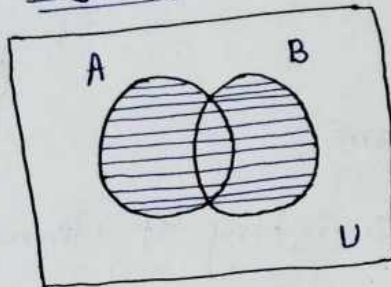
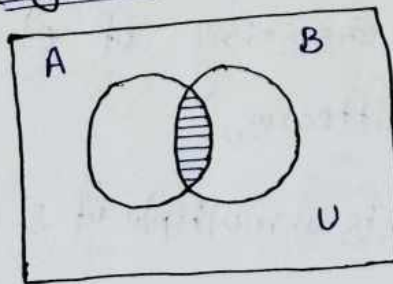
$$= 10 [12 + 95]$$

$$= 10 [107] = 10 \times 107 = 1070$$

$$\therefore \boxed{S_{20} = 1070}$$

Short Answer Questions.

- ① Draw the Venn diagram of (i) union of sets A and B.
(ii) Intersection of sets A and B.

Sol:-(i) union of sets A and B:-Venn Diagram:- $A \cup B =$ (ii) Intersection of sets A and B:-Venn Diagram:- $A \cap B =$ 

- ② write the principle of Inclusion-Exclusion for A, B, C sets.

Ans:-principle of Inclusion-Exclusion for A, B, C sets:-

1. If A, B are two sets then, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

2. If A, B, C are three sets then,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C).$$

3. $n(A - B) = n(A) - n(A \cap B)$.

③ write the given set in the set builder form $\{a, e, i, o, u\}$.

(ii) $A = \{3, 6, 9, 12, 15\}$.

Set Builder form:-

Sol:-

(i) The set A consisting of elements a, e, i, o, u .

can be written as,

$$A = \{x : x \text{ is a vowel in English alphabets}\}.$$

(ii) The set A consisting of elements $3, 6, 9, 12, 15$.

can be written as,

$$A = \{x : x \text{ is a multiple of 3 and lies between 3 and 15}\}.$$

④ Define finite and Infinite sets.

Ans
Finite set:- A set with finite number of elements in it is called a finite set.

Example:- 1. The set of months in a year.
 2. The set of vowels in English alphabets.
 3. $C = \{1, 2, 3, 4\}$.

Infinite set:- An Infinite set is a set which contains infinite number of elements.

Example:- 1. If $A =$ a set of Integers.
 2. If $B = \{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots\}$.

⑤ Definition of Cartesian product $A \times B$:-

Let A and B be two sets. then the Cartesian product of A, B is denoted by $A \times B$ and defined as set of all ordered pairs of A, B where $a \in A$ and $b \in B$.

$$\text{now, } A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Q6

Define singleton set and give example.

Ans

Definition of singleton set:-

A set which has only one element is called a singleton set.

Ex:- 1. $S = \{\omega\}$.

2. $\{\phi\}$ is a singleton set containing one element ϕ .

3. $\{0\}$ is a singleton set containing one element 0.

Q7

Represent the following set in Roostex form $\{x^2 - 3x + 2 = 0\}$.

Sol:-

Given that, $A = \{x : x^2 - 3x + 2 = 0\}$.

$$x^2 - 3x + 2 = 0$$

$$x^2 - 2x - x + 2 = 0$$

$$x(x-2) - 1(x-2) = 0$$

$$(x-2)(x-1) = 0$$

$$x = 2 \text{ and } x = 1.$$

$\therefore A = \{1, 2\}$ which is the representation of the given set A in tabular (or) Roostex form.

8

Define subsets of a set (or) power set.

Ag:-

Definition of subsets of a set (or) power set:-

For a set A a collection of all subsets of A is called the power set of A . and it is denoted by $P(A)$.

$$\therefore P(A) = \{x \mid x \subseteq A\}.$$

If a set containing 'n' elements, then the number of subsets is 2^n .

Ex:- 1. set $A = \{2, 4, 5, 6, 7\}$ containing five elements

So that total number of its subsets $2^5 = 32$.

2. List all the subsets of the set $A = \{a, b, c\}$.

Sol:- The subsets are $\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$.
[$\because 2^3 = 8$].

3. If $A = \{0, 1, 2\}$ find all subsets of A .

Sol:- The subsets of A are $\phi, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{1, 2\}, \{0, 2\}, \{0, 1, 2\}$.
[$\because 2^3 = 8$].

Long Answer Questions.

①

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ prove distributive laws

Proof:-

Let x be any arbitrary element of the set

$A \cup (B \cap C)$ then,

$$x \in A \cup (B \cap C) \Rightarrow x \in A \text{ or } x \in (B \cap C)$$

$$\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$$

$$\Rightarrow x \in (A \cup B) \text{ and } x \in (A \cup C)$$

$$x \in A \cup (B \cap C) \Rightarrow x \in (A \cup B) \cap (A \cup C)$$

$$\text{Thus } A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \longrightarrow (1)$$

Conversely suppose x be any arbitrary element of the set $(A \cup B) \cap (A \cup C)$ then,

$$x \in (A \cup B) \cap (A \cup C) \Rightarrow x \in (A \cup B) \text{ and } x \in (A \cup C)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$$

$$\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C)$$

$$\Rightarrow x \in A \text{ or } x \in (B \cap C)$$

$$x \in (A \cup B) \cap (A \cup C) \Rightarrow x \in A \cup (B \cap C)$$

$$\text{Thus } (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C) \longrightarrow (2)$$

$\therefore$ from (1) and (2)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Hence proved.

- (2) Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{3, 6, 8, 12, 17, 18\}$,
 $C = \{2, 3, 5, 6, 12, 17\}$, Find (i) $A \cup B$, (ii) $A \cap B$,
(iii) $A \cup C$, (iv) $A - B$, (v) $B - A$.

Sol:-

Given that, $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{3, 6, 8, 12, 17, 18\}$.
and $C = \{2, 3, 5, 6, 12, 17\}$.

(i) To Find $A \cup B$:-

Now, $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{3, 6, 8, 12, 17, 18\}$.

Now, $A \cup B = \{1, 2, 3, 4, 5, 6\} \cup \{3, 6, 8, 12, 17, 18\}$.

$\therefore A \cup B = \{1, 2, 3, 4, 5, 6, 8, 12, 17, 18\}$.

(ii) To Find $A \cap B$:-

Now, $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{3, 6, 8, 12, 17, 18\}$.

Now, $A \cap B = \{3, 6\}$.

(iii) To Find AUC:-

$$\text{Now, } A = \{1, 2, 3, 4, 5, 6\} \text{ and } C = \{2, 3, 5, 6, 12, 17\}.$$

$$\text{Now, } A \cup C = \{1, 2, 3, 4, 5, 6, 12, 17\}.$$

(iv) To Find A-B:-

$$\text{Now, } A = \{1, 2, 3, 4, 5, 6\} \text{ and } B = \{3, 6, 8, 12, 17, 18\}.$$

$$\text{Now, } A - B = \{1, 2, 3, 4, 5, 6\} - \{3, 6, 8, 12, 17, 18\}.$$

$$A - B = \{1, 2, 4, 5\}.$$

(v) To Find B-A:-

$$\text{Now, } B = \{3, 6, 8, 12, 17, 18\} \text{ and } A = \{1, 2, 3, 4, 5, 6\}.$$

$$\text{Now, } B - A = \{3, 6, 8, 12, 17, 18\} - \{1, 2, 3, 4, 5, 6\}.$$

$$B - A = \{8, 12, 17, 18\}.$$

③

Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (or) Prove distributive law.

Proof:-

Let x be any arbitrary element of the set $A \cap (B \cup C)$

Then, $x \in A \cap (B \cup C) = x \in A \text{ and } x \in B \cup C.$

$$\Rightarrow x \in A \text{ and } (x \in B \text{ or } x \in C)$$

$$\Rightarrow (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$$

$$\Rightarrow x \in A \cap B \text{ or } x \in A \cap C.$$

$$\Rightarrow x \in (A \cap B) \cup (A \cap C).$$

$$\text{Thus, } A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C) \rightarrow \textcircled{1}$$

Conversely, let x be any arbitrary element of the set $(A \cap B) \cup (A \cap C)$ then,

$$x \in (A \cap B) \cup (A \cap C) \Rightarrow x \in (A \cap B) \text{ or } x \in (A \cap C)$$

$$\Rightarrow (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$$

$$\Rightarrow x \in A \text{ and } (x \in B \text{ or } x \in C)$$

$$\Rightarrow x \in A \cap (B \cup C).$$

$$\text{Thus, } (A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C) \rightarrow \textcircled{2}$$

From $\textcircled{1}$ and $\textcircled{2}$.

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Hence proved.

- ④ If $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $C = \{1, 2, 3, 4, 5\}$ Find (i) $A \times B$
 (ii) $C \times B$ (iii) $B \times B$ hence prove that $(C \times B) - (A \times B) = B \times B$.

Sol:-

Given that, $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $C = \{1, 2, 3, 4, 5\}$.

(i) To find $A \times B$:-

$$\text{Now, } A \times B = \{1, 2, 3\} \times \{4, 5\}.$$

$$\text{Now, } A \times B = \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)\}.$$

(ii) To find $C \times B$:

$$\text{Now, } C \times B = \{1,2,3,4,5\} \times \{4,5\}.$$

$$= \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)(4,4)(4,5)(5,4)(5,5)\}.$$

(iii) To find $B \times B$:

$$\text{Now } B \times B = \{4,5\} \times \{4,5\}$$

$$= \{(4,4)(4,5)(5,4)(5,5)\}.$$

To prove that $(C \times B) - (A \times B) = B \times B$:

$$\text{Now, } C \times B = \{1,2,3,4,5\} \times \{4,5\}$$

$$= \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)(4,4)(4,5)(5,4)(5,5)\}.$$

$$\text{Now, } A \times B = \{1,2,3\} \times \{4,5\}.$$

$$= \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)\}.$$

L.H.S.:-

$$\text{Consider, } (C \times B) - (A \times B) = \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)(4,4)(4,5)(5,4)(5,5)\} - \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)\}.$$

$$= \{(4,4) (4,5) (5,4) (5,5)\}.$$

$$= B \times B$$

$$= R \cdot H \cdot S.$$

$$\therefore \boxed{(C \times B) - (A \times B) = B \times B}$$

Hence Proved.

- (5) List the Symmetric difference $A \Delta B$ If $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{0, 2, 4, 6, 8\}$

$$A \Delta B = (A - B) \cup (B - A)$$

$$A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$B = \{0, 2, 4, 6, 8\}$$

$$A - B = \{1, 3, 5, 7, 9\}$$

$$B - A = \{\} \text{ or } \phi$$

$$A \Delta B = (A - B) \cup (B - A)$$

$$= \{1, 3, 5, 7, 9\} \cup \{\}$$

$$A \Delta B = \{1, 3, 5, 7, 9\}$$

- (6) Prove that $A \cup B = B \cup A$

Let x be any arbitrary element of the $A \cup B$, then

$$x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$$

$$\Rightarrow x \in B \text{ or } x \in A$$

$$x \in A \cup B \Rightarrow x \in B \cup A$$

$$\text{Thus } x \in A \cup B \Rightarrow x \in B \cup A$$

$$A \cup B \subseteq B \cup A \text{ --- (1)}$$

Conversely Suppose that x be any arbitrary element of the set $B \cup A$ then

$$x \in B \cup A \Rightarrow x \in B \text{ or } x \in A$$

$$\Rightarrow x \in A \text{ or } x \in B$$

$$\Rightarrow x \in A \cup B$$

$$x \in B \cup A \Rightarrow x \in A \cup B \text{ --- (2)}$$

$$B \cup A \subseteq A \cup B \text{ --- (3)}$$

from eq ① & ② we have

$$A \cup B = B \cup A$$

- ⑦ If $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 5, 6, 7, 8\}$, $B = \{0, 1, 6, 7\}$
 $C = \{1, 2, 3, 5, 8\}$, verify 1) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
2) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Sol:-

Given that, $A = \{1, 5, 6, 7, 8\}$ and $B = \{0, 1, 6, 7\}$ and
 $C = \{1, 2, 3, 5, 8\}$.

To verify (1) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Consider, $B \cap C = \{0, 1, 6, 7\} \cap \{1, 2, 3, 5, 8\}$

$$B \cap C = \{1\}$$

Again consider,

$$\begin{aligned} A \cup (B \cap C) &= \{1, 5, 6, 7, 8\} \cup \{1\} \\ &= \{1, 5, 6, 7, 8\}. \end{aligned}$$

$\therefore$ L.H.S.:-

$$A \cup (B \cap C) = \{1, 5, 6, 7, 8\}.$$

Again,

$$\begin{aligned} \text{Consider, } A \cup B &= \{1, 5, 6, 7, 8\} \cup \{0, 1, 6, 7\} \\ &= \{0, 1, 5, 6, 7, 8\}. \end{aligned}$$

$$\begin{aligned} \text{and, } A \cup C &= \{1, 5, 6, 7, 8\} \cup \{1, 2, 3, 5, 8\} \\ &= \{1, 2, 3, 5, 6, 7, 8\} \end{aligned}$$

Again, Consider, $(A \cup B) \cap (A \cup C) = \{0, 1, 5, 6, 7, 8\} \cap \{1, 2, 3, 5, 6, 7, 8\}$
 $= \{1, 5, 6, 7, 8\}.$

R.H.S.:-

$$\therefore (A \cup B) \cap (A \cup C) = \{1, 5, 6, 7, 8\}.$$

$$\therefore A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\therefore \text{L.H.S} = \text{R.H.S.}$$

$\therefore$ verified.
Hence proved.

To verify that (2) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$

$$\text{Now, } B \cup C = \{0, 1, 6, 7\} \cup \{1, 2, 3, 5, 8\}$$

$$= \{0, 1, 2, 3, 5, 6, 7, 8\}.$$

$$\text{Consider, } A \cap (B \cup C) = \{1, 5, 6, 7, 8\} \cap \{0, 1, 2, 3, 5, 6, 7, 8\}$$

$$= \{1, 5, 6, 7, 8\}.$$

L.H.S.:-

$$A \cap (B \cup C) = \{1, 5, 6, 7, 8\}.$$

Again, Consider,

$$A \cap B = \{1, 5, 6, 7, 8\} \cap \{0, 1, 6, 7\}$$

$$= \{1, 6, 7\}.$$

$$\text{and } A \cap C = \{1, 5, 6, 7, 8\} \cap \{1, 2, 3, 5, 8\}$$

$$= \{1, 5, 8\}.$$

Again consider,

$$(A \cap B) \cup (A \cap C) = \{1, 6, 7\} \cup \{1, 5, 8\} \\ = \{1, 5, 6, 7, 8\}.$$

R.H.S:-

$$(A \cap B) \cup (A \cap C) = \{1, 5, 6, 7, 8\}.$$

$$\therefore A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

$$\therefore L.H.S = R.H.S.$$

$\therefore$ verified.

Hence Proved.

⑧

In a college of 120 students, 70 students are in drama club, 50 students are in music club and 30 are in both clubs then, find. (i) students who are in either drama or music.

(ii) students who are in only drama.

(iii) students who are only in music.

(iv) students who are in neither club.

Sol:-

(i) Given, $n(A) = 70$, $n(B) = 50$, $n(A \cap B) = 30$.

$$\text{Now, } n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$= 70 + 50 - 30$$

$$= 120 - 30 = 90.$$

$$\therefore n(A \cup B) = 90$$

$\therefore$ students who are in either drama or music = 90

$$(ii) \quad n(A - B) = n(A) - n(A \cap B)$$

$$= 70 - 30$$

$$\therefore n(A - B) = 40$$

$\therefore$ students who are in drama only = 40

$$(iii) \quad n(B - A) = n(B) - n(A \cap B)$$

$$= 50 - 30$$

$$= 20$$

$\therefore$ students who are in music only = 20

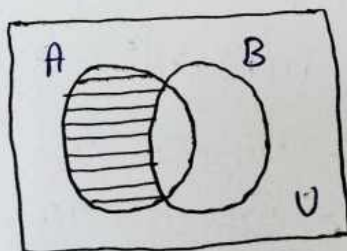
$$(iv) \quad \text{students who are in neither club} = 120 - 90 = 30$$

9 Draw the following Venn diagrams, (i) $A - B$ (ii) $A \cup (B \cap C)$
(iii) $(A \cap B) \cup (A \cap C)$ (iv) $(A \cup B) \cap (A \cup C)$ (v) $A \cap (B \cap C)$

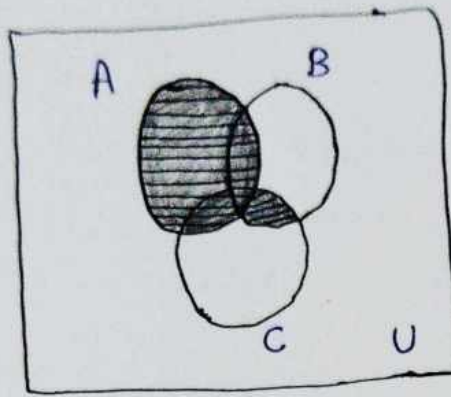
Sol:-

Draw the Venn diagrams:-

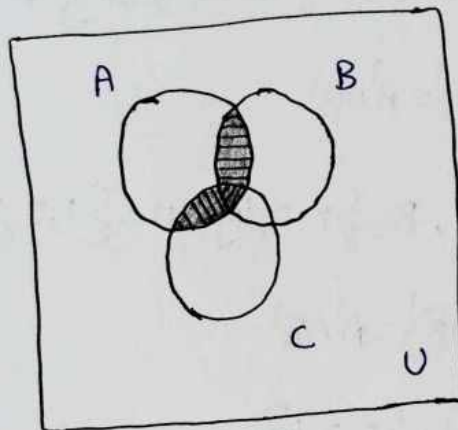
(i) $A - B$



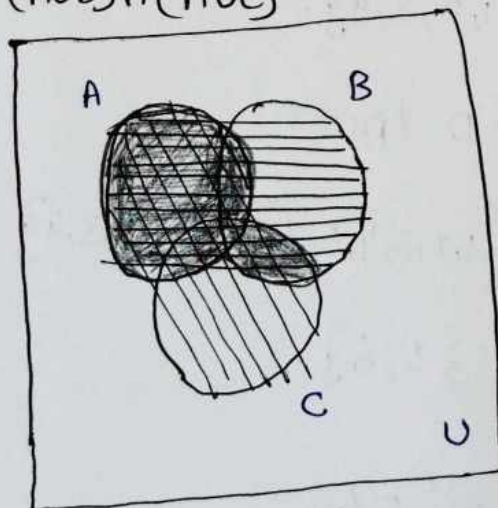
(ii) $A \cup (B \cap C)$



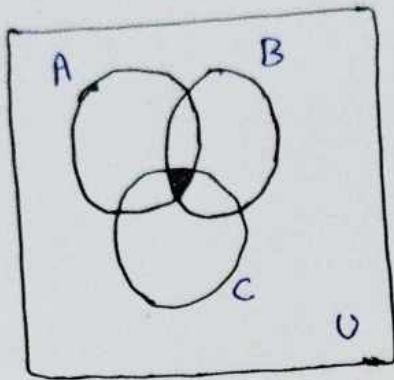
(iii) $(A \cap B) \cup (A \cap C)$



(iv) $(A \cup B) \cap (A \cup C)$



(V) $A \cap (B \cap C)$



10

If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and $U = \{1, 2, 3, 4, 5, 6\}$ verify that

1) $(A \cup B)' = A' \cap B'$ 2) $(A \cap B)' = A' \cup B'$.

Sol:-

Given that, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $U = \{1, 2, 3, 4, 5, 6\}$.

(i) To verify that $(A \cup B)' = A' \cap B'$.

$$\begin{aligned} \text{Now, } A \cup B &= \{1, 2, 3\} \cup \{2, 3, 4\} \\ &= \{1, 2, 3, 4\}. \end{aligned}$$

$$\begin{aligned} \text{Now, } (A \cup B)' &= U - (A \cup B) \\ &= \{1, 2, 3, 4, 5, 6\} - \{1, 2, 3, 4\} \\ &= \{5, 6\}. \end{aligned}$$

L.H.S:-

$$(A \cup B)' = \{5, 6\}$$

now, $A = \{1, 2, 3\}$.

now, $A' = U - A$.

$$= \{1, 2, 3, 4, 5, 6\} - \{1, 2, 3\}$$

$$= \{4, 5, 6\}.$$

and $B = \{2, 3, 4\}$.

now, $B' = U - B$

$$= \{1, 2, 3, 4, 5, 6\} - \{2, 3, 4\}$$

$$= \{1, 5, 6\}.$$

Consider, $A' \cap B' = \{4, 5, 6\} \cap \{1, 5, 6\}$

$$= \{5, 6\}.$$

R.H.S:-

$$A' \cap B' = \{5, 6\}.$$

$$\therefore (A \cup B)' = A' \cap B'$$

$$\therefore L.H.S = R.H.S$$

$\therefore$ verified.

(ii) To verify that $(A \cap B)' = A' \cup B'$.

$$\text{now, } A \cap B = \{1, 2, 3\} \cap \{2, 3, 4\}$$

$$A \cap B = \{2, 3\}.$$

$$\text{Now, } (A \cap B)' = U - (A \cap B) \\ = \{1, 2, 3, 4, 5, 6\} - \{2, 3\}$$

L.H.S. - $(A \cap B)' = \{1, 4, 5, 6\}$

$$\text{Now, } A' = U - A \\ = \{1, 2, 3, 4, 5, 6\} - \{1, 2, 3\} \\ = \{4, 5, 6\}$$

$$\text{and } B' = U - B \\ = \{1, 2, 3, 4, 5, 6\} - \{2, 3, 4\} \\ = \{1, 5, 6\}$$

$$\text{Now, } A' \cup B' = \{4, 5, 6\} \cup \{1, 5, 6\} \\ = \{1, 4, 5, 6\}$$

R.H.S. - $A' \cup B' = \{1, 4, 5, 6\}$

$$\therefore (A \cap B)' = A' \cup B'$$

$$\therefore \text{L.H.S.} = \text{R.H.S.}$$

$\therefore$ verified.